D5 Risk Assessment

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Approach

Our group adopted the use of a risk register using Software Engineering[1] as a guideline which was applied to our risk management strategy. Risks were identified collaboratively during the requirements and plans to mitigate these risks were reviewed at our bi-weekly meetings. At each of the meetings we would evaluate each risk based on the likelihood and impact using a simplified 3-point scale: Low, Medium and High and help to see which risks needed to be prioritised.

The justification for our risk management plan was to accurately: *identify, analyse, mitigate, monitor and dedicate ownership of each* unique risk that happened during the development of our project.

Process

1. Identification - Thinking of different types of risks that may come up in the project as well as adding new risks which appeared throughout the project.
2. Analysis - What is the likelihood of the occurrence and how it will impact the project.
3. Mitigation - What can be done to minimise the impact of the risk and help prevent this happening again in the future.
4. Monitoring - Bi-weekly review of the risk register and adding new risks if any were identified.
5. Ownership - Each risk was assigned to a team / team member to oversee.

After this process was completed, the risks were continued to be monitored by each unique owner and discussed at the bi-weekly meetings to see how progress was being made on each of the risks.

This seemed like the most efficient process to allow us to keep track of all of the risks as the risks were clearly displayed and each one assigned to someone so all risks stayed accounted for. We could have used other methods like SWOT (strengths, weaknesses, opportunities and threats) to categorize the risks for our project. However, we felt like this would not be optimal for a project of a small scale and we felt like creating a risk register would allow us to track the risks more efficiently.

Risk Register

1. Risk ID - A numerical ID given to each risk which helps to track risks throughout the project.
2. Type - What type of risk it is. It is split into three distinct categories: Business, Product and Project.
3. Description - A short write up of what each risk is about.
4. Likelihood - The chances of one of these risks happening or recurring throughout the project.
5. Severity - How much of an impact can be done each risk will make to the project.
6. Mitigation - Solutions that can be done to either prevent the risk happening again or minimise the effect.
7. Owner - Who will be responsible for each risk.

Risk ID	Type	Description	Likelihood	Severity	Mitigation	Owner

R1	Product and Project	Version control conflicts when multiple members edit at the same time	L	M	Effective communication amongst the group to prevent conflicting versions	Owner of the PR

R2	Project	Members absent for various reasons	H	H	Reassignment of tasks and roles + work to be done on a shared drive to track progress	Sam
R3	Project	Underestimating time required for key and core game mechanics	H	H	Setting weekly specific goals, progress is discussed at meetings	Anexa
R5	Project	JAR does not run consistently across Windows / Mac / Linux	L	H	Avoided OS specific dependencies	Wasif & Hamza
R6	Business	Misinterpreting client expectations for events/difficulty amongst the group	L	M	Documented the client meeting and clarified with each other	Emma
R7	Project	Poor team communication resulting in missed meetings and unclear tasks	M	M	Bi-weekly set meetings + table showing what everyone's tasks and roles are	Leo
R8	Product	Gameplay bugs affect core mechanics	H	M	Run tests automatically on every PR.	Sam
R9	Product	Low code quality / inconsistent style	M	M	Coding style standardised and checked automatically by Checkstyle	Sam

R11	Project	Risks were not identified at the start of the project	M	M	Risks recorded and identified throughout the project	Emma
R13	Product	Performance issues across different hardware	L	L	Tested across multiple different hardwares	Wasif & Hamza
R14	Project	Documentation inconsistent with the code (Not updated UML Diagrams)	M	H	Documentation done in tandem with implementation	Leo
R15	Project	Game files lost due to corruption	L	H	Regular copies of the game saved	Sam
R16	Project	Members falling behind due to prioritising other academic workload / other commitments	H	L	Good communication and workload shared accordingly	Anexa
R17	Product	Dependencies and 3rd party libraries become incompatible / unavailable	M	H	Alternative options considered and downloading copies of the dependencies	Wasif

Bibliography

[1] Sommerville, Ian, “Software Engineering,” Pearson Education, 2015